Assignment No. 8

1. **Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).**

```
KD3_rutuja_96226>SELECT * FROM orders WHERE cnum = (SELECT cnum from customers
WHERE cname ='Cisneros');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.011 sec)
```

2. **Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.**

```
KD3_rutuja_96226>SELECT c.cname, c.rating, o.amt FROM customers c INNER JOIN orders
 o ON c.cnum=o.cnum WHERE amt > (SELECT AVG(amt) from orders);
+---------+--------+---------+
| cname   | rating | amt     |
+---------+--------+---------+
| Liu     |    200 | 5160.45 |
| Clemens |    100 | 9891.88 |
| Clemens |    100 | 4723.00 |
+---------+--------+---------+
3 rows in set (0.025 sec)
```

3. **Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.**

```
KD3_rutuja_96226>SELECT snum, SUM(amt) from orders GROUP BY snum HAVING SUM(amt)>
(SELECT MAX(amt) from orders);
+------+----------+
| snum | SUM(amt) |
+------+----------+
| 1001 | 15382.07 |
+------+----------+
1 row in set (0.048 sec)
```

4. **Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).**

```
KD3_rutuja_96226>select * from customers where rating >= ANY (SELECT rating FROM
customers WHERE snum = (SELECT snum FROM salespeople WHERE sname = 'serres'));
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.886 sec)
```

5. **Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).**

```
KD3_rutuja_96226>SELECT * FROM salespeople WHERE city != ALL(SELECT city FROM
customers WHERE salespeople.snum = customers.snum);
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.663 sec)
```

6. **Write a query that selects all orders for amounts greater than any for the customers in London.**

```
KD3_rutuja_96226>SELECT * FROM orders WHERE amt >ANY (SELECT amt from orders where
cnum in(select cnum from customers where city='london'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.227 sec)
```

7. Extract all the orders of Motika.

```
KD3_rutuja_96226>SELECT * FROM orders where snum = (SELECT snum from salespeople
WHERE sname = 'Motika');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.021 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
KD3_rutuja_96226>select * from orders where snum in (select snum from customers
 where city = 'london');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.123 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
KD3_rutuja_96226>SELECT sname, snum from salespeople WHERE snum IN
(SELECT snum FROM customers GROUP BY snum HAVING count(snum) > 1);
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.264 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
KD3_rutuja_96226>SELECT s.snum, s.sname, s.city, (SELECT COUNT(*) FROM customers c WHERE
c.snum = s.snum) as cnt FROM salespeople s WHERE (SELECT COUNT(*) FROM customers c WHERE
c.snum = s.snum) > 1;
+------+--------+----------+------+
| snum | sname  | city     | cnt  |
+------+--------+----------+------+
| 1001 | Peel   | London   |    2 |
| 1002 | Serres | San Jose |    2 |
+------+--------+----------+------+
2 rows in set (0.772 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
KD3_rutuja_96226>SELECT * from customers WHERE rating >= ANY(SELECT MAX(rating) from customers
where city = 'Rome');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.008 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from '1990-10-04'.

```
KD3_rutuja_96226>SELECT * FROM orders WHERE amt > (SELECT MIN(amt) from orders WHERE odate = '1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.793 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose .

```
KD3_rutuja_96226>SELECT * from orders where amt < any(SELECT amt from orders where cnum
 IN(SELECT cnum from customers where city='San Jose'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.030 sec)
```

14. Select those customers whose rating are higher than every customer in London.

```
KD3_rutuja_96226>SELECT * FROM customers WHERE rating > ALL (SELECT rating FROM customers
WHERE cnum IN (SELECT cnum FROM customers WHERE city = 'London'));
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.105 sec)
```